THE FIVE STRANDS OF MATHEMATICAL PROFICIENCY

Mathematical proficiency is developed through the interweaving of five essential strands. Together they form a strong, flexible foundation for mathematical thinking and learning.



1 CONCEPTUAL UNDERSTANDING

Comprehension of mathematical concepts, operations, and relations.

2 PROCEDURAL FLUENCY

Skill in carrying out procedures flexibly, accurately, efficiently, and appropriately.

3 STRATEGIC COMPETENCE

Ability to formulate, represent, and solve mathematical problems.

4 ADAPTIVE REASONING

Capacity for logical thought, reflection, explanation, and justification.

5 PRODUCTIVE DISPOSITION

Habitual inclination to see mathematics as sensible, useful, and worthwhile, coupled with a belief in diligence and one's own efficacy.

INTERWOVEN. INTERDEPENDENT. ESSENTIAL.

When all five strands are developed together, learners are empowered to think mathematically and thrive.

Adapted from National Research Council (2001) Adding it Up I Poster design and adaptation © 2026 Luis Miguel Fernández, Ph.D.